Functions & Validation Gap Report

Code-based status report for the inspection, customer estimate approval, tool tracking, and parts-waste workflows.

REVIEW DATE	SCOPE	OVERALL STATUS
October 31, 2026	**Current implemented workflow code**	**Core flows exist; follow-up gaps remain**

Executive Summary

The requested core workflows are present in the current code: initial inspection, token-based customer decisions, tool checkout/return, maintenance, and wasted-parts stock recording. The main remaining work is completing the surrounding operational lifecycle, tightening duplicate/re-entry rules, and adding customer communications. This report focuses on those workflows; it is not a full security certification or a line-by-line audit of every module.

1. Functions Already Present

- **Initial inspection:** staff can record findings and recommended work for a checked-in appointment, attach up to five photos, and view inspection history and photos.
- **Customer estimate approval:** creates a single-use link that expires after seven days; customer can approve all, approve selected items, decline, or request changes. Stale revisions and invalid item selections are rejected.
- **Tool checkout and return:** records the assigned employee, optional work order, checkout/return condition, and audit events. Damaged returns open a repair-maintenance record; missing returns mark the tool inactive.
- **Tool maintenance:** staff can start and complete a maintenance record, record cost and condition, set a next-due date, and update calibration dates for calibration work.
- **Wasted parts:** the parts ledger can record waste against approved/reserved scope, update stock/reservation quantities, and keep waste separate from customer-billable installed parts.

2. Functions Still Needed

Priority	Function to add	Expected behavior
High	Customer link delivery and reminders	Email/SMS the approval link, record delivery outcome, and provide a resend action. Keep the current expiry and one-use behavior.
High	Changes-requested revision workflow	Show the customer note to staff, create a revised estimate, invalidate the old link, and issue a fresh link.
High	Inspection correction / re-inspection lifecycle	Support a controlled correction or re-inspection with reason, actor, timestamp, and a clear link to the prior inspection.
Medium	Approval-link revoke/resend controls	Allow staff to revoke a still-valid link or issue a replacement without waiting for expiry.
Medium	Missing-tool recovery / disposition	Provide a workflow to mark a missing tool recovered, retired, or written off, with reason and audit history.
Medium	Inspection media lifecycle	Add retention/deletion handling for stored photos and ensure storage is included in backup/restore procedures.

Note: the customer approval link is currently a bearer link. Anyone with the URL can submit a decision while it is valid; a separate identity check is not part of this flow.

Code References

- Inspection and photo handling: `Controllers/OwnerController.InitialInspection.cs`
- Approval token and decision validation: `Controllers/OwnerController.RepairWorkflow.cs` and `Controllers/CustomerEstimateApprovalController.cs`
- Tool checkout, return, and maintenance: `Controllers/OwnerController.ToolTransactions.cs`
- Form constraints: `Models/RepairWorkflowModels.cs`; inspection UI: `Views/Owner/Appointments/Appointments.cshtml`.

Basis: source-code inspection in the current workspace on October 31, 2026. This is not a full security certification; run HTTP/database smoke tests after changes.

3. Validations Already Present

Area	Current validation / safeguard
Initial inspection	Requires a checked-in appointment with customer and vehicle; findings are required and must be at least 10 characters; maximum 3,000 characters.
Inspection photos	Allows up to 5 files, max 5 MB each; accepts JPG/PNG/WebP and checks file signatures against declared type.
Customer decision	Decision limited to approved/partial/declined/changes requested; requires valid token, pending current revision, and unexpired link; rejects duplicate/foreign line IDs and invalid selection combinations.
Approval submission	POST uses antiforgery validation and serializable database transaction; successful decision consumes the link and is audited.
Tool checkout	Requires active, non-archived, non-deleted tool; safe condition; calibration not overdue; no open checkout; active employee; and active work order/assignment when linked.
Tool return	Requires an open checkout and existing tool; accepted condition values are Excellent, Good, Fair, Damaged, or Missing; notes are required and capped at 1,000 characters.
Tool maintenance	Prevents open checkout or duplicate open maintenance; completion requires an allowed safe condition and rejects a next-due date in the past; cost has a 0 to 10,000,000 range.
Wasted parts	Requires approved/reserved scope and tracks inventory movement separately from installed/billable parts; stock and reservation are adjusted through the ledger.

4. Validation Gaps / Risks

Priority	Gap	Recommended validation
High	Repeat initial inspections are not blocked or explicitly marked as re-inspections.	Reject a second inspection by default, or require a re-inspection reason and link it to the previous inspection. Prevent inspection changes after work has advanced unless reopened.
High	Inspection photo persistence is not in the same database transaction as inspection metadata.	Write files to temporary storage, persist metadata and inspection transactionally, then finalize files; clean up orphaned files after failures.
Medium	Maintenance type is free text and calibration due date is optional.	Use a defined maintenance-type list; require a next-due date for calibration work and validate it against the configured schedule.
Medium	No endpoint or permission-protected screen resolves a Missing tool status.	Require an audited resolution action; do not allow a missing tool to become available through ordinary edit fields.

Implementation Order

1. Add duplicate/re-inspection rules and transactional photo handling. 2. Add customer link email/resend/revoke and the changes-requested revision loop. 3. Add audited missing-tool disposition. 4. Tighten calibration-specific maintenance validation.